Chapter 7: Business Intelligence

Preparation For AI: From Raw Data To Reliable Models

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Business Intelligence

Data preparation is not finished when a file is written.

Chapter 7 asks:

```
How do humans inspect and discuss prepared data?
```

The answer is SQL exploration plus dashboards.

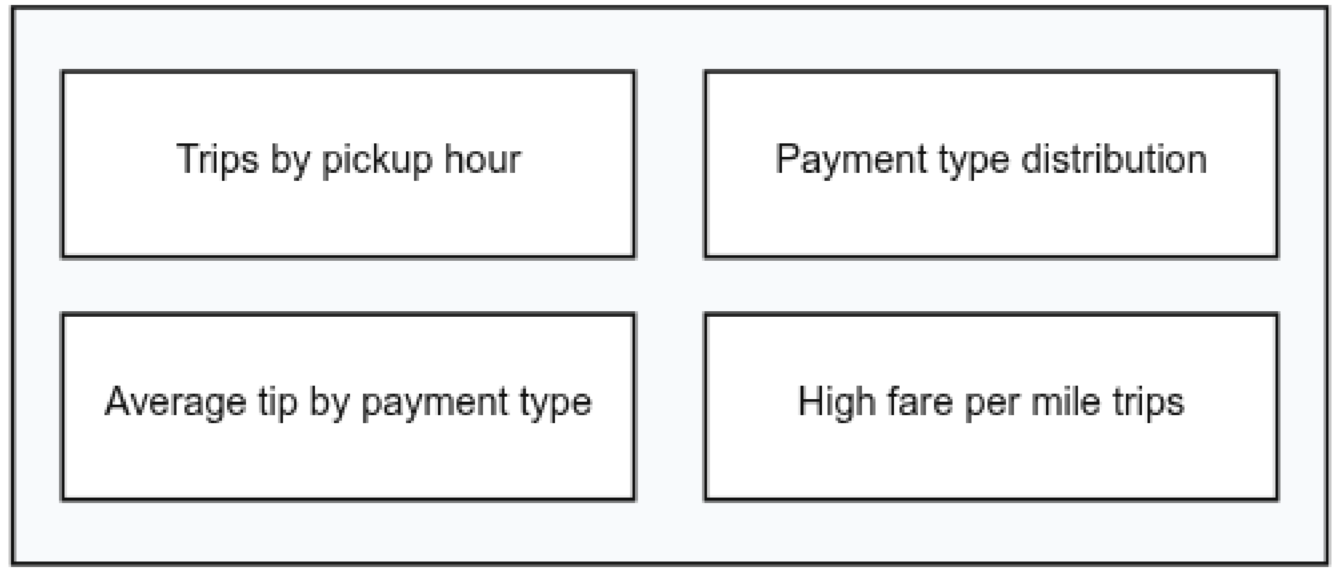
Prepared Data Needs A Human Interface

Logs, Parquet files, Iceberg snapshots, and Airflow runs are useful evidence.

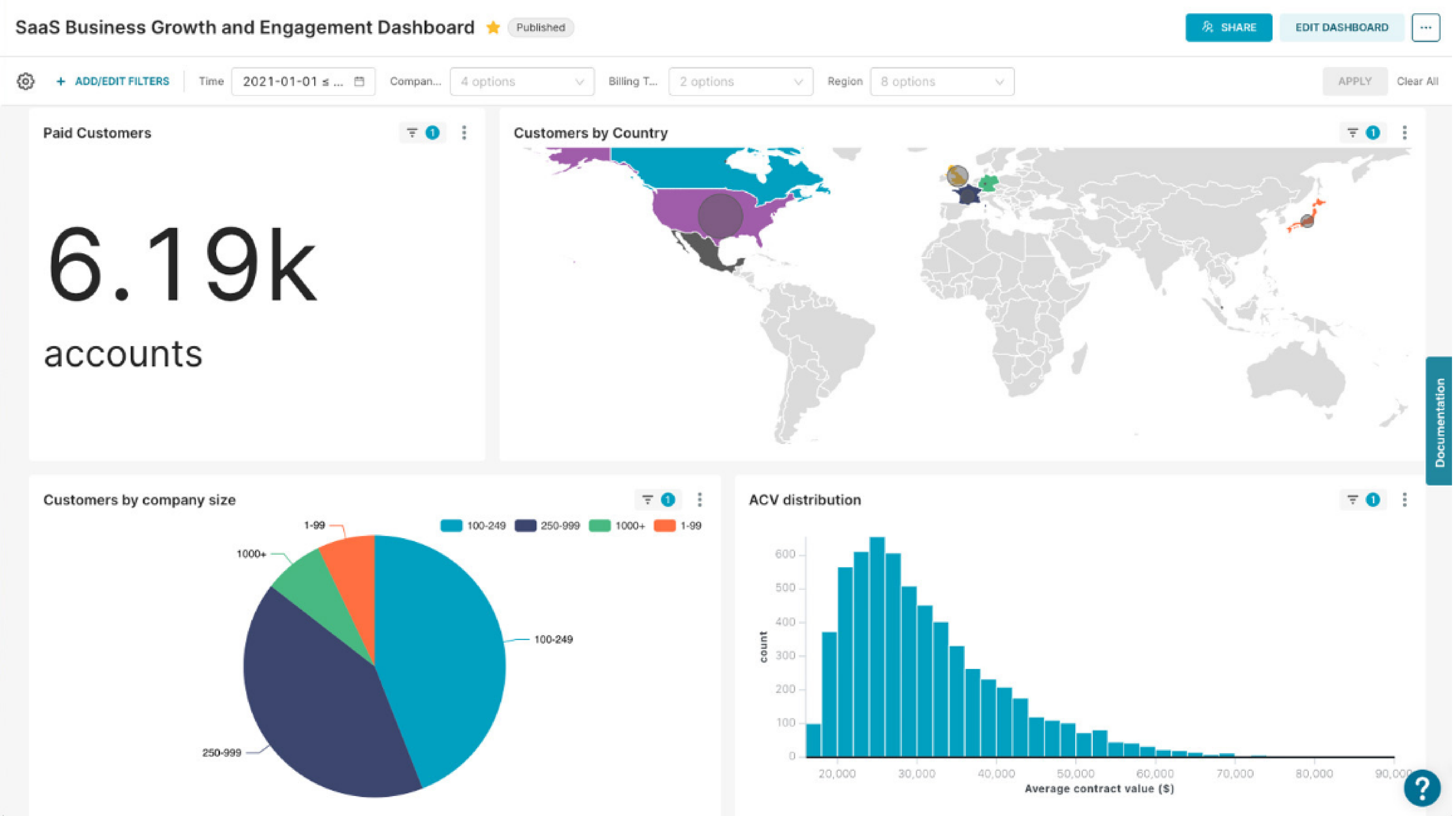
But wider teams need:

- SQL questions
- charts
- filters
- dashboards
- shared discussion

A useful dashboard is a set of related questions



Superset UI



SaaS Business Growth and Engagement Dashboard
Published
SHARE
EDIT DASHBOARD
+ ADD/EDIT FILTERS
Time
2021-01-01 ≤ ...
Compan...
4 options
Billing T...
2 options
Region
8 options
APPLY
Clear All
Paid Customers
6.19k
accounts
Customers by Country
Documentation
Customers by company size
100-249
250-999
1000+
1-99
ACV distribution
count
Average contract value ($)

BI Is Visibility, Not Preparation

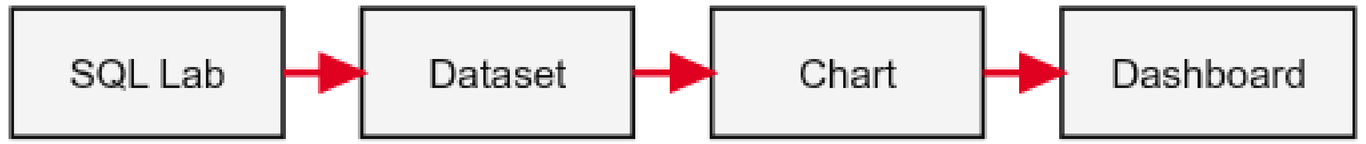
BI should sit after preparation:

```
prepared table -> SQL question -> chart -> discussion
```

Do not rebuild cleaning logic inside every chart.

If the data is wrong, fix the pipeline, not just the dashboard.

Use SQL first, then build reusable views



If a chart looks wrong, check the SQL result.

BI Changes The Audience

Business Intelligence helps engineers talk with:

- domain experts
- analysts
- managers
- researchers
- operations teams

The goal is shared inspection, not decoration.

Questions BI Can Answer

Useful BI questions:

- how many records arrived?
- which categories dominate?
- which values look suspicious?
- did a metric change?
- which filters change the conclusion?

These are team questions.

Superset Is Not The Lake

Superset is:

- SQL Lab
- chart builder
- dashboard builder
- filters
- sharing interface

It is not the storage layer or transformation engine.

The BI Stack

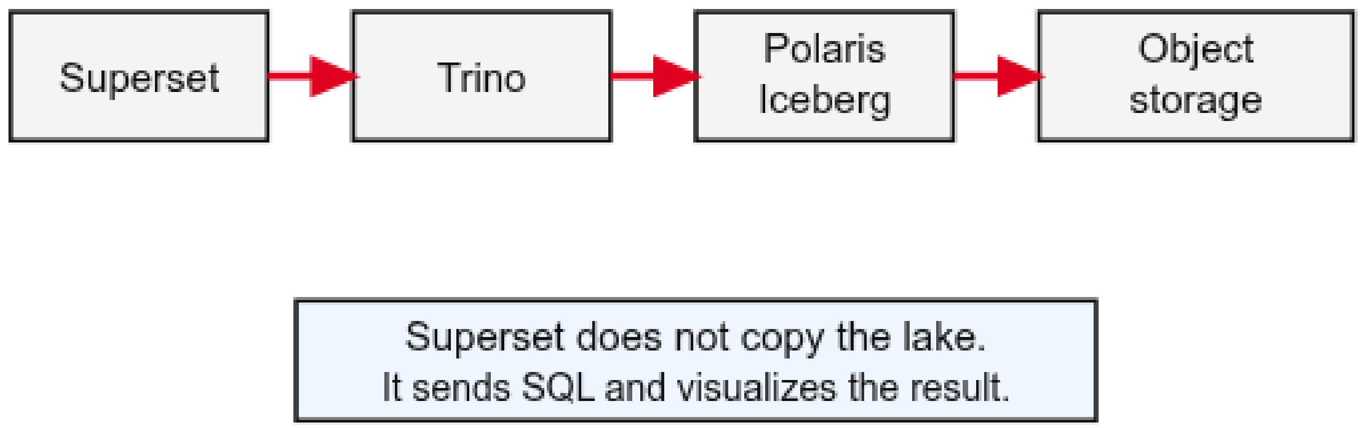
In this chapter:

```
Superset -> Trino -> Polaris / Iceberg -> object storage
```

Each layer has a separate responsibility.

The dashboard does not copy the data lake.

BI sits at the human end of the lakehouse path



Dimensions And Metrics

A dashboard is usually not a table dump.

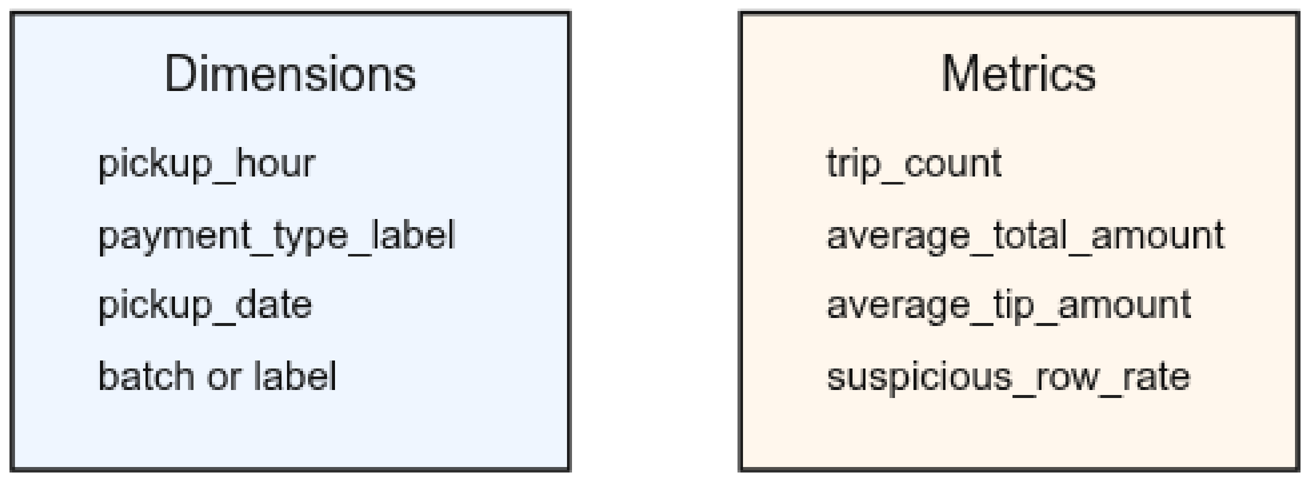
It is a set of questions expressed through:

- dimensions: what we group or filter by
- metrics: what we calculate

Dimensions give charts their shape.

Metrics give charts their values.

Dashboards are built from dimensions and metrics



One Chart, One Question

A useful chart should make one question easier to answer.

Not:

```
make a graph because we can
```

But:

```
which payment type behaves differently?
```

SQL Lab First

SQL Lab is the sanity-check layer.

Use it to inspect:

- columns
- row counts
- categories
- suspicious values
- aggregation results

Then build charts.

Superset Dataset

In Superset, a dataset means:

- table or SQL source
- columns
- temporal columns
- dimensions
- metrics
- filterable fields

It is Superset’s bridge from SQL to chart builder.

Dashboards Are Conversations

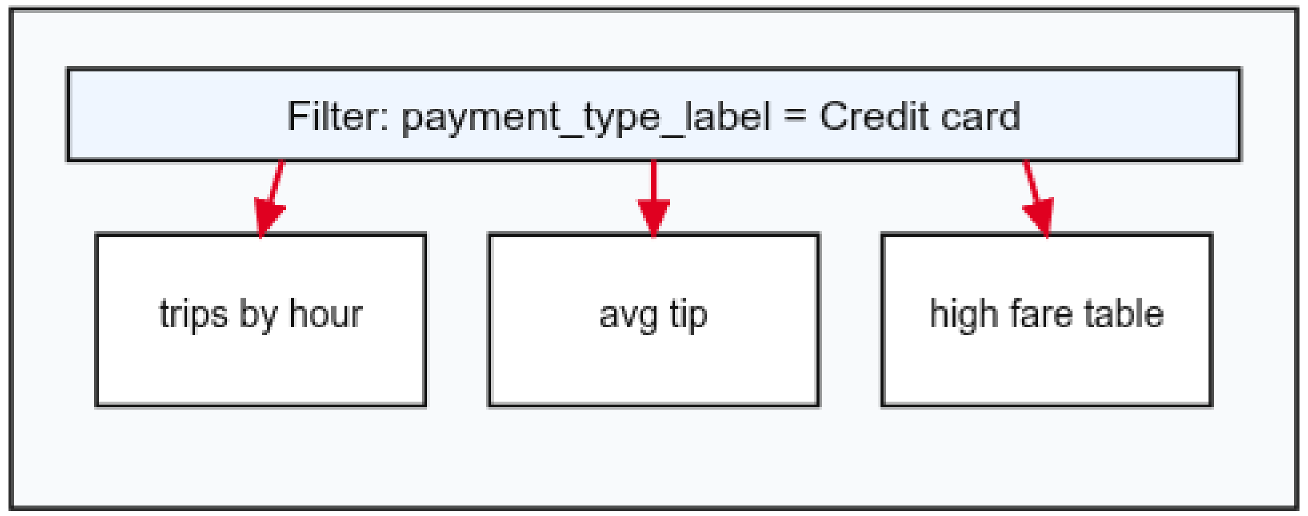
A dashboard collects related charts and filters.

Useful dashboard views should answer related questions.

Filters let users ask follow-up questions.

The dashboard should create a conversation, not a poster.

Filters turn static charts into exploration



Suspicious Is Not A Label

High fare per mile is a clue.

It does not automatically mean fraud or bad data.

Use context:

- distance
- duration
- payment type
- total amount
- pickup time

BI helps start the investigation.

What BI Is For

Good fit:

- queryable tabular data
- reusable charts and dashboards
- filters for follow-up questions
- team discussion around prepared data

Poor fit:

- raw multimedia inspection
- model artifact tracking
- training-run comparison

Other tools handle those better.

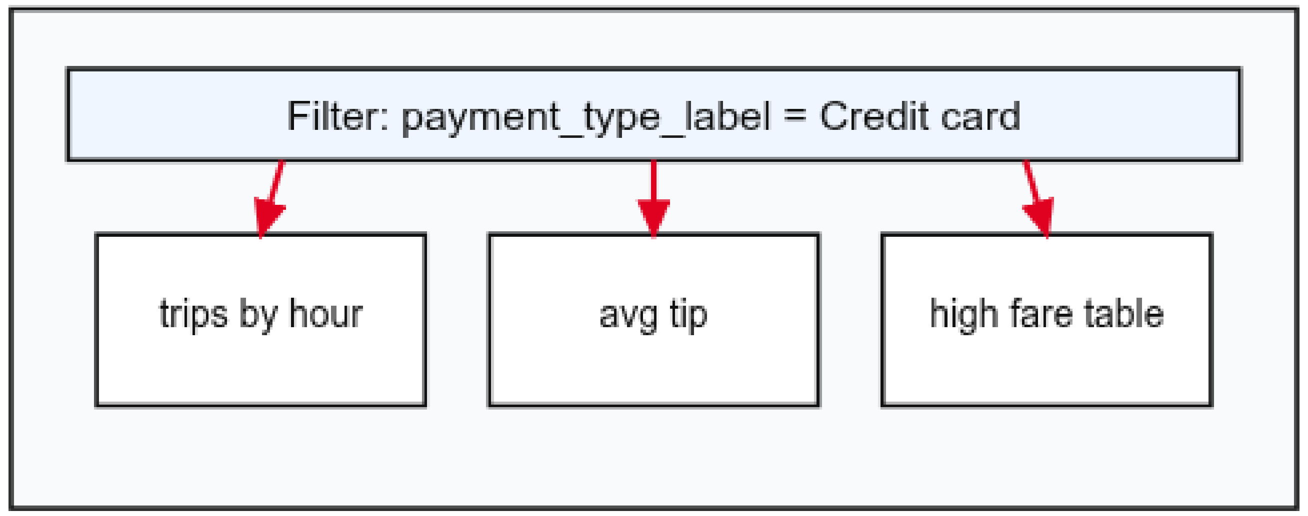
Lab Orientation

The shared table is:

```
business_intelligence_lab.taxi_trips
```

Students use SQL Lab to inspect it, then build charts and a filtered dashboard.

Filters turn static charts into exploration



The Investigation Question

The dashboard should help answer:

```
When are taxi trips most common,
and which payment types behave differently?
```

The answer should live in the dashboard interaction.

BI And AI Preparation

BI can reveal:

- broken preparation
- surprising category balance
- suspicious values
- changed distributions
- unclear definitions

Those issues matter before training models.

Business Intelligence

Prepared data should be visible.

Superset gives humans SQL, charts, dashboards, and filters.

Trino queries the lakehouse table.

BI turns prepared data into **shared questions**.

Next: track model training and serve predictions.